Standard Operating Procedure

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Title:	Time-Course Bioreactor Sampling and Viability Analysis		
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Trellis Bioworks   Project: CHO-K1 Fed-Batch Optimization   Generated: May 17, 2026

1.0 Purpose

Standardise the time-course sampling and in-process viability analysis of CHO-K1 fed-batch cultures across the 2 L Sartorius Biostat platform so that inter-operator and inter-batch results are directly comparable.

2.0 Scope

Applies to all GLP fed-batch development runs performed in Suite 2B on the Biostat A 2 L vessels (BR-201 through BR-204). Does not cover seed-train shake-flask sampling.

3.0 Procedure: Time-Course Bioreactor Sampling and Viability Analysis

CRITICAL REQUIREMENT: All sampling MUST occur within ±5 min of the nominal time point. Deviations >5 min require an in-process deviation (IPD) report before the next time point.

3.1 Hour 0 (T=0) — Inoculation Sampling

Reference time point. Sample drawn immediately after inoculation establishes baseline viability and metabolite concentration.

Figure 1: Inoculation sampling port (post-CIP).

Annotated photograph of the bioreactor sampling port showing the sterile septum and integrated luer fitting.

Time Target	Action	Expected Output / Log
T=0 min	Aseptically draw 5 mL via sterile sampling port into pre-labelled 15 mL conical.	Sample S-0 logged in eBR; cap torque verified.
T+10 min	Perform trypan-blue viability count (Cellometer K2, 1:1 dilution).	Viable cell density (cells/mL); viability %.
T+15 min	Submit retained aliquot to in-process QC (pH, glucose).	pH, glucose (g/L) recorded; sample chain-of-custody signed.

3.2 Hour 3 (T=3) — Foam Growth Check

Inspection only — no draw unless foam height exceeds the documented limit.

Figure 2: Acceptable vs. excessive foam.

Side-by-side comparison of acceptable headspace foam (10 mm) and the rejection threshold.

Time Target	Action	Expected Output / Log
T=3 h	Visually inspect headspace foam height; compare to Figure 2.	PASS / FAIL noted in eBR; foam height (mm).
T=3 h +5 min	If FAIL: add 0.5 mL sterile antifoam-C via dedicated addition port and re-inspect.	Antifoam lot, volume, and post-add foam height logged.

3.3 Hour 6 (T=6) — Feed Initiation Sampling

Sample drawn immediately prior to bolus feed addition; establishes the pre-feed metabolic profile.

Time Target	Action	Expected Output / Log
T=6 h	Aseptically draw 10 mL into pre-labelled tube; split into viability and metabolite aliquots.	Sample S-6 logged; aliquots routed to bench + QC.
T=6 h +20 min	Initiate bolus feed (Feed-A, 50 mL) via peristaltic pump P-02.	Feed start time, pump RPM, and totaliser reading.